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A Stochastic Volatility Model With Markov Switching

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This article presents a new way of modeling time-varying volatility. We generalize the usual stochastic volatility models to encompass regime-switching properties. The unobserved state variables are governed by a first-order Markov process. Bayesian estimators are constructed by Gibbs sampling. High-, medium- and low-volatility states are identified for the Standard and Poor's 500 weekly return data. Persistence in volatility is explained by the persistence in the low- and the medium-volatility states. The high-volatility regime is able to capture the 1987 crash and overlap considerably with four U.S. economic recession periods.

KEY WORDS: ARCH model; Bayesian inference; Data augmentation; Gibbs sampling; Monte Carlo Markov chain.

It is well known that the volatility of financial series tends to change over time. The autoregressive conditional heteroscedasticity (ARCH) model introduced by Engle (1982) provided a breakthrough in the modeling and estimation of time-varying conditional variance. Since then, many generalizations of the ARCH model have emerged. For comprehensive surveys of the models in the ARCH family, one can refer to Bollerslev, Chou, and Kroner (1992) and Bollerslev, Engle, and Nelson (1994). Alternatively, Taylor (1986) argued that the volatility process should be driven by economic forces rather than the movement of prices. The class of models that is formulated under this kind of belief is usually referred to as the *stochastic volatility* (SV) model. There have been many attempts in developing models for stochastic volatility. See, for example, Clark (1973), Tauchen and Pitts (1983), Harris (1987), and Gallant, Hsieh, and Tauchen (1991). Moreover, SV models give a discrete-time approximation for the diffusion processes used in option pricing. See, for example, Wiggins (1987), Hull and White (1987a,b), and Heston (1993), in which the Black–Scholes option-pricing formula was generalized to incorporate stochastic volatility. Comprehensive reviews of the stochastic volatility models can be found in the work of Taylor (1994), Ghysels, Harvey, and Renault (in press), and Shephard (1996).

The aim of this article is to present a new way to explain volatility persistence in the SV framework. One common finding in the ARCH and the SV literature is the persistence of volatility, especially in high-frequency data. For example, Chou (1988) documented that persistence of shocks to the stock-market volatility was high in the U.S. stock market during 1962–1985. Similar results can be observed in the work of French, Schwert, and Stambaugh (1987). Poon and Taylor (1992) remarked that the persistence of stock volatility is self-evident in some high-frequency U.K. data. So, Lam, and Li (1997) demonstrated strong evidence of volatility persistence in the Taiwan stock market.

Lamoureux and Lastrapes (1990) suggested that the apparent persistence of the variance may be overestimated because the possible structural shift in the model had not been taken into account. In the same article, the authors pointed out that means of identifying occasional switching in the parameter values, like the Markov switching model of Hamilton (1988, 1989), may provide more appropriate modeling of volatility. In this regard, Hamilton and Susmel (1994) proposed the Markov switching ARCH (SWARCH) model, which is an ARCH specification with changes over regimes. The scale of the conditional variance is allowed for structural changes according to an unobservable Markov process. In the application to the New York Stock Exchange data, most of the persistence of the shock is accounted for by the persistence in either regime. Cai (1994) applied a similar idea by incorporating both the switching and the ARCH models. The article noted that the volatility of treasury bills becomes much less persistent when allowing for a switching regime in some parameters.

We generalize the SV model to incorporate regime-switching properties. As in the SWARCH model, the parameter determining the level of the logarithm of volatility is allowed occasional discrete shifts. The switching dynamic is governed by a first-order Markov process. In the Markov switching SV (MSSV) model, the scale of the volatility is allowed to change according to a Markov process. Therefore, the MSSV model simultaneously captures the usual

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changing behavior of volatility due to economic forces as well as the sudden discrete shift in volatility due to sudden abnormal events.

Parameter estimation in SV models can be performed by the generalized method of moments (GMM) type of estimators (Taylor 1982, 1986, 1994; Andersen 1994; Andersen and Sørensen 1996). The extensive Monte Carlo study of Andersen and Sørensen (1996) provided guidance on GMM estimation of SV models. Nelson (1988), Harvey and Shephard (1996), Harvey, Ruiz, and Shephard (1994), and So, Li, and Lam (1997) approximated an SV model by a Gaussian state-space model and derived the quasi-maximum likelihood (QML) estimators for the unknown parameters. A new approach that should be brought to attention is the Markov-chain Monte Carlo (MCMC) technique suggested by Shephard (1993) and Jacquier, Polson, and Rossi (1994). Samples from the joint posterior density of the parameters are obtained through Gibbs sampling (Tierney 1994). Any function of the parameters can be estimated by the Gibbs sample. In this article, we adopt the Bayesian approach in which the MCMC and data augmentation (Tanner and Wong 1987) are combined for parameter estimation.

This article is organized as follows. Section 1 defines the MSSV model. Section 2 gives a brief review of the Gibbs-sampling methodology. Section 3 shows how to apply the MCMC technique to obtain samples from the joint posterior distribution of the unknown parameters. Section 4 describes methods for smoothing and predicting the volatility. Section 5 constitutes a simulation study to demonstrate the MCMC technique. The methodology is applied to the Standard and Poor's (S & P) 500 data in Section 6, with conclusions in Section 7.

1. THE MSSV MODEL

1.1 The SV Model for Financial Time Series

Let y_t denote a financial time series. The SV model as formulated by Shephard (1996) is defined as

$$y_t = \sqrt{h_t} u_t, \tag{1}$$

$$\log h_{t+1} = \alpha + \phi \log h_t + \eta_t, \tag{2}$$

where u_t and η_t are independent normal random variables with zero mean and variances 1 and σ^2, respectively. The drift coefficient of y_t is assumed to be 0. Under the model formulation, the variable y_t is the product of two stochastically independent variables $\sqrt{h_t}$ and u_t. Asymmetric volatility can be captured by allowing contemporaneous correlation between u_t and η_t (Harvey and Shephard 1996).

In terms of state-space representation, Equation (2) is the transition equation in which $\log h_t$ is the state variable. From Equation (1), we have the corresponding measurement equation

$$\log y_t^2 = \log h_t + \xi_t, \tag{3}$$

where $\xi_t = \log u_t^2$. In the QML estimation procedure, the density of ξ_t is approximated by a normal density with mean -1.2704 and variance $\pi^2/2$ (Harvey et al. 1994).

1.2 A New Model

Suppose that s_t is an unobserved discrete random variable with domain $\{1, 2, \ldots, K\}$. Assuming a K-state first-order Markov process, we can write the transition probability matrix as

$$\begin{pmatrix} p_{11} & p_{12} & \cdots & p_{1K} \\ p_{21} & p_{22} & \cdots & p_{2K} \\ \cdot & \cdot & & \cdot \\ \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & & \cdot \\ p_{K1} & p_{K2} & \cdots & p_{KK} \end{pmatrix},$$

where $p_{ij} = \Pr(s_t = j | s_{t-1} = i)$ with $\sum_{j=1}^{K} p_{ij} = 1$ for $i = 1, \ldots, K$. The state variable s_t defines a particular regime for the parameter values. For example, $s_t = 1$ corresponds to the first regime and $s_t = K$ corresponds to the last regime. Similar to the idea of Hamilton (1989) for conditional mean in autoregressive processes, Equation (2) generalizes as follows:

$$\log h_{t+1} = \alpha_{s_{t+1}} + \phi \log h_t + \eta_t, \tag{4}$$

where

$$\alpha_{s_t} = \gamma_1 + \sum_{j=2}^{K} \gamma_j I_{jt} \tag{5}$$

and I_{jt} is an indicator variable that is equal to 1 when s_t is greater than or equal to j. Equations (1), (4), and (5) together define the K-state Markov switching stochastic volatility or MSSV(K) model. The following additional assumptions are made:

1. The coefficient ϕ should lie inside the unit circle.
2. Each regime should correspond to at least one time point.
3. All $\gamma_j (j = 2, \ldots, K)$ are negative.

The assumptions are similar to those of Albert and Chib (1993). Assumption 1 guarantees the stationarity of $\log h_t$ given the state s_t. Assumption 2 is necessary; otherwise, some γ_j's will be unidentified. The parameterization in (5) can be written as $\alpha_{s_t} = \sum_{j=1}^{s_t} \gamma_j$. Therefore, assumption 3 implies that the larger the value of s_t, the smaller the level. Therefore, the first regime defines the highest-level state, whereas the last regime defines the lowest-level state. For example, when $K = 3$, $s_t = 1$, 2, and 3 correspond to the high-volatility, the medium-volatility, and the low-volatility states, respectively. The MSSV(K) model introduced here allows different α in different regimes that can capture occasional discrete shifts in volatility.

2. GIBBS SAMPLING AND DATA AUGMENTATION

Gibbs sampling, a Monte Carlo integration technique using a Markovian updating scheme, was formally introduced by Geman and Geman (1984) in the context of image reconstruction. It is essentially a modification of the Metropolis algorithm of Metropolis, Rosenbluth, Rosenbluth, Teller, and Teller (1953). The substitution algorithm for the data-augmentation scheme of Tanner and Wong (1987) is closely

related to Gibbs sampling. Gelfand and Smith (1990) and Gelfand, Hills, Poon, and Smith (1990) illustrated a wide range of applications in linear models. Gibbs sampling has been used in estimation of time series. See, for example, Carlin, Polson, and Stoffer (1992), Chib (1993), Carter and Kohn (1994), Shephard (1993, 1994), and Jacquier et al. (1994), who gave the first application for the analysis of stochastic volatility.

The data-augmentation technique is particularly useful in models having latent variables. In the MSSV(K) model, let θ be the parameter vector, Y_t be the information up to time t—that is, $Y_t = (y_1, \ldots, y_t)'$—and $H_t = (h_1, \ldots, h_t)'$, $S_t = (s_1, \ldots, s_t)'$ be the latent variable vectors. As shown by Jacquier et al. (1994), we augment the parameter vector θ by the latent vectors H_n and S_n. Now the parameter space is enlarged with $\omega = (H_n', S_n', \theta')'$ as an element. The decomposition of the joint posterior density according to Bayes's theorem,

$$f(H_n, S_n, \theta|Y_n) \propto f(Y_n|H_n)f(H_n|S_n, \theta)f(S_n|\theta)f(\theta), \quad (6)$$

allows us to sample from the full conditional posterior distributions. The details will be given in Section 3.

3. IMPLEMENTATION OF THE MCMC TECHNIQUE

To generate the MCMC from the Gibbs sampler, the crucial point is to sample from the full conditional distributions, $F(\omega_i|\omega_{-i}, Y_n)$. Following (6),

$$f(\omega_i|\omega_{-i}, Y_n) \propto f(Y_n|H_n)f(H_n|S_n, \theta)f(S_n|\theta)f(\theta), \quad (7)$$

where

$$f(Y_n|H_n) \propto \prod_{t=1}^{n} h_t^{-1/2} \exp\left(-\frac{1}{2}\frac{y_t^2}{h_t}\right),$$

$$f(H_n|S_n, \theta) \propto \sigma^{-n}\sqrt{1-\phi^2}\left(\prod_{t=1}^{n} h_t^{-1}\right) \times \exp\left\{-\frac{1}{2\sigma^2}\left[\sum_{t=2}^{n}(\log h_t - \alpha_{s_t} - \phi \log h_{t-1})^2 + (1-\phi^2)\left(\log h_1 - \frac{\alpha_{s_1}}{1-\phi}\right)^2\right]\right\},$$

$$f(S_n|\theta) = \prod_{t=2}^{n} p_{s_{t-1}s_t} \times \pi_{s_1},$$

$$\pi_i = \Pr(s_1 = i), \quad i = 1, \ldots, K,$$

and $f(\theta)$ depends on the choice of the prior distribution for the unknown parameter θ. We assume that

$$f(\theta) = f(\sigma^2)f(\phi)\prod_{i=1}^{K}[f(\gamma_i)f(p_{i1}, \ldots, p_{iK})f(\pi_i)]. \quad (8)$$

In practice, the Gibbs sampling of drawing successively from the full conditional distributions is iterated for $M+N$ times. The first M burn-in iterations are discarded. The last N iterates are taken to be an approximate sample from the joint posterior distribution. The full conditional distributions for the model parameters are given in the Appendix. Simulations of the latent variables follow.

3.1 Drawing $s_t(t = 1, \ldots, n)$

We have at least two possible ways to simulate the latent data S_n. The single-move sampler draws the sequence of s_t's from the full conditional distributions $f(s_t|S_{-t}, Y_n, H_n, \theta)$, where S_{-t} denotes the set of S_n without s_t. This method was used by Albert and Chib (1993) in their two-regime Markov switching autoregressive model. Although these procedures were applied to some observable time series, they can be easily generalized to our situation, in which the state variable s_t determines the drift of the latent variable h_t.

Here we adopt another strategy that simulates S_n from the joint distribution $f(S_n|Y_n, H_n)$. The dependence on θ is suppressed for notational convenience. This algorithm is called the multimove sampler. It was described by Carter and Kohn (1994) and Chib (1995, 1996) in different contexts. It is expected that the multimove sampler for S_n can lead to faster convergence in the MCMC. Consider the decomposition of $f(S_n|Y_n, H_n)$ as

$$f(S_n|Y_n, H_n) = f(s_n|Y_n, H_n)\prod_{t=1}^{n-1} f(s_t|Y_n, H_n, S^{t+1}), \quad (9)$$

where $S^t = (s_t, \ldots, s_n)'$. Equation (9) provides a route to draw S_n in a single block using the method of composition, given that we know how to sample from $f(s_t|Y_n, H_n, S^{t+1}), t = 1, \ldots, n-1$, and $f(s_n|Y_n, H_n)$. Now, for $t = 1, \ldots, n-1$,

$$\begin{aligned} f(s_t|Y_n, H_n, S^{t+1}) &= f(s_t|Y_t, H_t, s_{t+1}) \\ &\propto f(s_{t+1}|s_t, Y_t, H_t)f(s_t|Y_t, H_t) \\ &= f(s_{t+1}|s_t)f(s_t|Y_t, H_t). \end{aligned} \quad (10)$$

The expressions in (10) imply that the conditional probability $f(s_t|Y_t, H_t)$ is the crucial element. Analogous to the Kalman filter for a general state-space model, Carter and Kohn (1994) and Chib (1996) developed a discrete filter for the evaluation of $f(s_t|Y_t, H_t)$. Using the filter, desired samples from $f(S_n|Y_n, H_n)$ are generated via (9) and (10).

3.2 Drawing $h_t(t = 1, \ldots, n)$

To complete the construction of the MCMC, we have to sample the h_t's from suitable conditional distributions. Jacquier et al. (1994) showed that the h_t's can be drawn from the set of univariate densities,

$$f(h_t|h_{t-1}, h_{t+1}, \theta, y_t), \qquad t = 1, \ldots, n. \quad (11)$$

The conditional density reduces to $f(h_1|h_2, \theta, y_1)$ and $f(h_n|h_{n-1}, \theta, y_n)$, respectively, for $t = 1$ or n. Arguing according to Jacquier et al. (1994), the conditional densities in (11) take an unusual form that is essentially a product of normal and lognormal density functions. The authors in the article created a Metropolis subchain within the MCMC

constructed by the Gibbs sampler to sample from the nonstandard distribution. Geweke (1994) suggested that rejection sampling can be applied to draw h_t. The method is feasible if the SV model is adequate for the data; otherwise, a huge rejection rate may result in the sampling of h_t. The aforementioned approaches draw h_t separately without considering the correlation between the latent variables. Shephard (1994) emphasized that the multimove Gibbs sampler, which draws the time series of latent variables simultaneously, is much more efficient than the ordinary single-move sampler. Shephard and Kim (1994) and Kim, Shephard, and Chib (in press) documented the superiority of the multimove sampler to the usual Gibbs sampler when applied to the stochastic volatility model.

We adopt the method proposed by Shephard (1994), which formulates the MSSV model in the partial non-Gaussian state-space form. The trick of the approach is to approximate the $\log \chi_1^2$ distribution of ξ_t by a mixture of normal distributions; that is,

$$f(\xi_t) \approx \sum_{i=1}^{P} q_i f(\xi_t | \rho_t = i),$$

where

$$\xi_t | \rho_t = i \sim N(\mu_i - 1.2704, \tau_i^2), \qquad q_i = \Pr(\rho_t = i).$$

Following Kim et al. (in press), $P = 7$, and the parameters of the mixture distributions are also as given by Kim et al. (in press).

Given the ρ_t's as specified under the normal mixture approximation, the MSSV model can be written in an ordinary linear state-space form (Harvey 1993); that is,

$$\begin{aligned} \log y_t^2 &= \log h_t + \varepsilon_t \\ \log h_{t+1} &= \alpha_{s_{t+1}} + \phi \log h_t + \eta_t, \end{aligned}$$

where $\varepsilon_t | \rho_t \sim N(\mu_t', \tau_t^2), \mu_t' = \mu_{\rho_t} - 1.2704$, and $\tau_t^2 = \tau_{\rho_t}^2$. The recent results of De Jong and Shephard (1995) are employed in this article. Instead of sampling the state variables $\log h_t$'s directly from $f(\log h_1, \ldots, \log h_n | \rho_1, \ldots, \rho_n, \theta, S_n, Y_n)$, the disturbances η_t's are drawn from their multivariate posterior distribution $f(\eta_1, \ldots, \eta_n | \rho_1, \ldots, \rho_n, \theta, S_n, Y_n)$. The disturbance smoother of De Jong and Shephard (1995) has some advantages—for example, avoiding possible degeneracies and reducing storage requirement in the state sampler.

Finally, sampling of ρ_t is relatively easy to perform because of the discrete support and the independence of the time series ρ_t. The full conditional density of ρ_t is proportional to

$$\begin{aligned} &f(\log y_t^2 | \rho_t, h_t) f(\rho_t) \\ &\quad \propto \frac{1}{\tau_{\rho_t}} \exp\left[-\frac{1}{2\tau_{\rho_t}^2} (\log y_t^2 - \log h_t - \mu_{\rho_t} + 1.2704)^2 \right] f(\rho_t) \end{aligned}$$

(Shephard 1994). We draw ρ_t separately by using random numbers from the uniform distribution over [0, 1] (Ripley 1987).

Table 1. Results From the Simulated Data

Parameter	True values	Posterior mean	Posterior S.D.	Posterior 90% interval
γ_1	−1	−.916	.231	(−1.307, −.520)
γ_2	−1	−1.071	.167	(−1.353, −.808)
γ_3	−3	−3.005	.332	(−3.574, −2.438)
σ^2	.2	.314	.160	(.101, .607)
ϕ	.5	.513	.059	(.409, .605)
p_{11}	.9	.877	.049	(.790, .940)
p_{12}	0	.026	.040	(.000, .106)
p_{13}	.1	.097	.039	(.036, .165)
p_{21}	0	.027	.031	(.000, .088)
p_{22}	.95	.941	.040	(.870, .982)
p_{23}	.05	.032	.027	(.001, .084)
p_{31}	.05	.079	.031	(.033, .134)
p_{32}	.05	.040	.024	(.006, .085)
p_{33}	.9	.881	.034	(.828, .929)

Table 2. Summary Statistics for the S & P 500 Data

	r_t	r_t^2	$\log r_t^2$
Mean	.00103	.000474	−9.16
S. D.	.0217	.00230	2.12
Skewness	−1.51	30.4	−.995
Kurtosis	22.0	1,050	1.45
LB(24)	39.68 (25.18)	16.17 (227.96)	148.26 (110.32)
	Autocorrelations		
Lags			
1	.014 (.029)	.020 (.173)	.087 (.075)
2	−.042 (−.028)	.062 (.097)	.078 (.065)
3	.037 (.029)	.023 (.034)	.052 (.037)
4	.008 (.031)	.007 (.089)	.071 (.069)
5	−.049 (−.056)	.020 (.202)	.097 (.093)

NOTE: Kurtosis refers to excess kurtosis. LB(24) is the Ljung–Box statistics using first 24 autocorrelations. Values in parentheses are computed using data before the 1987 crash.

4. SMOOTHING AND PREDICTION OF THE VOLATILITY

Forecasting is an important aspect in the modeling of volatility. At the end of the Gibbs iterations, a sample from the joint posterior distribution $F(H_n, S_n, \theta | Y_n)$, denoted by $(h_1^{(i)}, \ldots, h_n^{(i)}, s_1^{(i)}, \ldots, s_n^{(i)}, \theta^{(i)}), i = M+1, \ldots, M+N$, is readily obtained. Smoothed estimates of $h_t (t = 1, \ldots, n)$ are estimates of h_t derived from the marginal posterior distribution $F(h_t | Y_n)$. A natural choice is the marginal posterior expectation $E(h_t | Y_n)$, which can be estimated by the sample mean,

$$\frac{1}{N} \sum_{i=M+1}^{M+N} h_t^{(i)}.$$

We predict future h_t as did Albert and Chib (1993).

Table 3. Results From the SV Fitting to the S & P 500 Data

Parameter	Posterior mean	Posterior S. D.	Posterior 90% interval
α	−.292	.095	(−.459, −.140)
σ^2	.045	.013	(.026, .068)
ϕ	.963	.012	(.943, .982)
Half-life (weeks)	21.381	10.577	(11.807, 38.792)

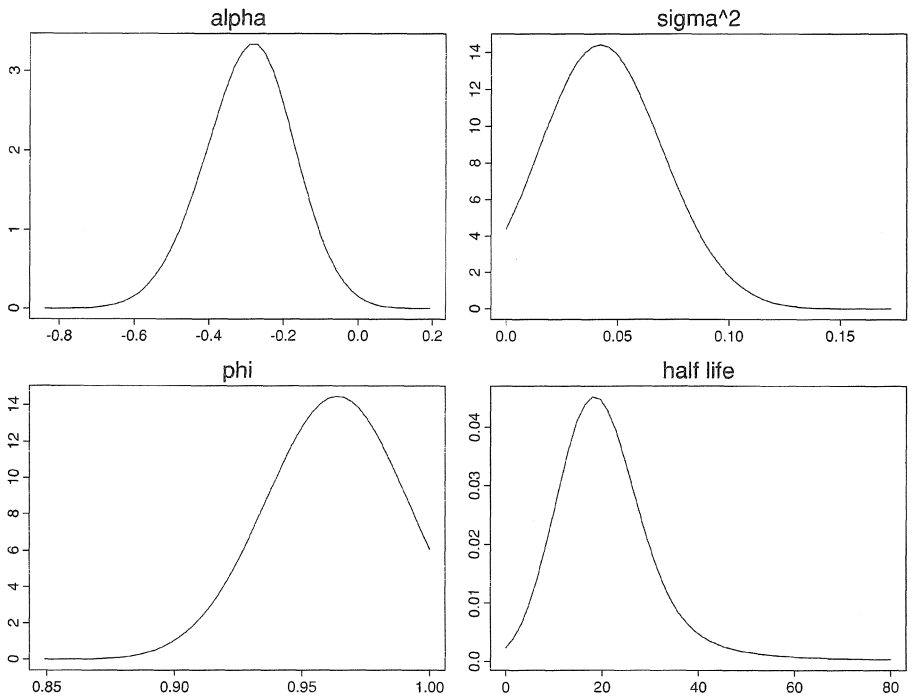


Figure 1. Posterior Densities for α, σ^2, ϕ, and the Half-life (weeks) for the S & P 500 Data in the SV Fitting.

5. SIMULATION STUDY

We now apply the methods described in the previous sections to a simulated example. The model we consider in the simulation exercise is

$$y_t = \sqrt{h_t}u_t,$$

$$\log h_{t+1} = \alpha_{s_{t+1}} + .5 \log h_t + \eta_t,$$

where $\eta_t \sim N(0, .2)$, and

$$\alpha_{s_t} = \begin{cases} -1, & \text{if } s_t = 1 \\ -2, & \text{if } s_t = 2 \\ -5, & \text{if } s_t = 3 \end{cases}$$

Table 4. Results From the MSSV Fitting to the S & P 500 Data

Parameter	Posterior mean	Posterior S. D.	Posterior 90% interval
γ_1	−3.536	1.300	(−5.325, −.264)
γ_2	−.764	.313	(−1.624, −.428)
γ_3	−.541	.167	(−.824, −.276)
σ^2	.074	.073	(.003, .217)
ϕ	.472	.158	(.234, .751)
p_{11}	.859	.202	(.176, .981)
p_{12}	.130	.188	(.015, .745)
p_{13}	.011	.024	(.000, .037)
p_{21}	.029	.018	(.006, .059)
p_{22}	.967	.020	(.931, .992)
p_{23}	.004	.005	(.000, .014)
p_{31}	.009	.010	(.000, .031)
p_{32}	.017	.015	(.001, .047)
p_{33}	.974	.015	(.946, .994)
$(1 - p_{11})^{-1}$	17.996	17.764	(1.215, 51.786)
$(1 - p_{22})^{-1}$	48.764	44.459	(14.496, 132.540)
$(1 - p_{33})^{-1}$	64.898	109.200	(18.395, 178.615)

The formulation implies that $\gamma_1 = -1, \gamma_2 = -1$, and $\gamma_3 = -3$. The state variables $s_1, \ldots, s_n$ are generated by a first-order Markov process with transition probability matrix

$$\begin{pmatrix} .9 & 0 & .1 \\ 0 & .95 & .05 \\ .05 & .05 & .9 \end{pmatrix}.$$

Most of the masses in the transition probability matrix are concentrated in the diagonal entries, implying high persistence in each regime. The measure of persistence is equal to .5. Indeed, all choices of parameters are coincident with the phenomena observed by Hamilton and Susmel (1994) when studying the weekly New York Stock Exchange index data by the SWARCH model.

A dataset of $n = 400$ observations was simulated from the model. Diffuse priors were used for the parameters $\gamma_1, \gamma_2, \gamma_3$, and ϕ. The prior mean of σ^2 was set at .1, and a large prior standard deviation of σ^2 was specified to indicate little information about σ^2. The first 10,000 burn-in iterations were discarded. A further 40,000 iterations were taken to be a sample from the joint posterior distributions of the unknown parameters as well as the latent variables s_t's and h_t's. Using the notation as in the previous sections, $M = 10,000$ and $N = 40,000$. The marginal posterior densities were constructed by keeping every 10th draw in the Gibbs sample. Using every 10th one in the full sample of 40,000 iterates better approximates a stream of independent observations. Therefore, information from the marginal posterior densities is extracted from a stream of 4,000 iterates instead of 40,000 iterates. Note that, if the standard error of the posterior moment estimates are required, one may use the robust variance estimation meth-

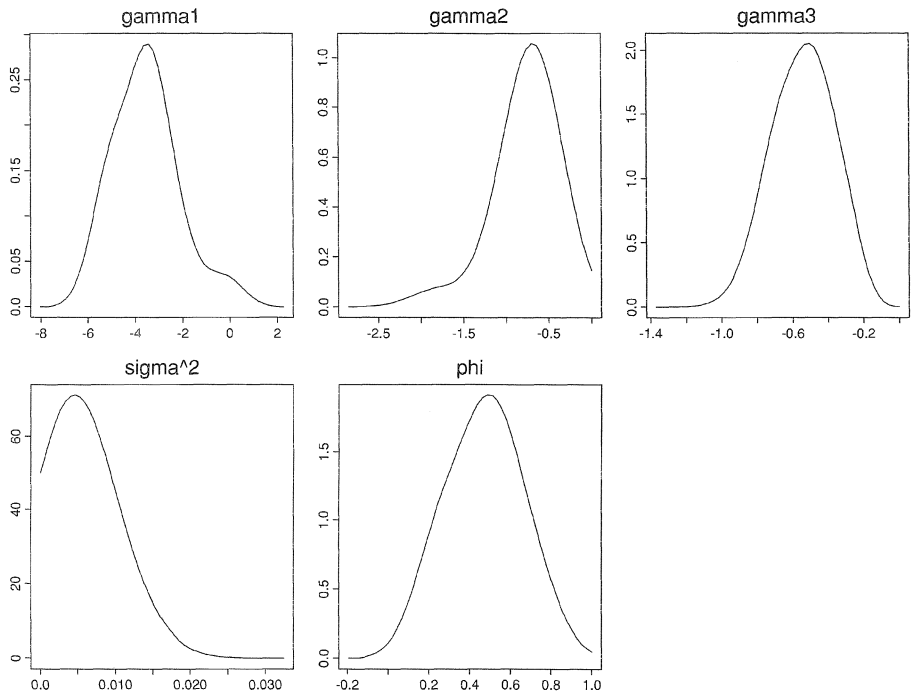


Figure 2. Posterior Densities for γ_1, γ_2, γ_3, σ^2, and ϕ for the S & P 500 Data in the MSSV(3) Fitting.

ods suggested by Andrews (1991) and Hansen (1992). The results of the Gibbs sampling are summarized in Table 1, page 247, where the posterior means, standard deviations, and 90% intervals are given. The posterior intervals are constructed so that 5% of the iterates are smaller than the lower limit and 5% of the iterates are greater than the upper limit. As a point estimate of the true parameter, all the posterior means are close to the corresponding true values. Moreover, all the posterior 90% intervals cover the true parameter values except p_{12} and p_{21}. The posterior means of p_{12} and p_{21} are small compared with their posterior standard deviations; also, the lower limits of their 90% intervals are close to 0. Thus, one may argue that these two entries in the transition probability matrix are essentially 0. Alternatively, one-sided intervals for these parameters could be considered. Overall, the marginal posterior densities give good descriptions of all the unknown parameters.

6. APPLICATION TO S & P 500 DATA

6.1 Data Description

The series we use for illustration is the S & P 500 index weekly return series from the first week of 1961 to the last week of 1987. Wednesday-to-Wednesday returns are generated. If the Wednesday index is not available, Thursday or Friday indexes are substituted. There are altogether 1,408 observations. Table 2, page 247, provides summary statistics for r_t, r_t^2, and $\log r_t^2$, where r_t is the S & P 500 return at time t.

Both series have excess kurtosis greater than 0. The autocorrelations and the Ljung–Box statistics for the transformed data r_t^2 are unexpectedly small. The 1987 crash may have had a strong influence here. For the sake of comparison, the sample autocorrelations, as well as the Ljung–Box statistics, are recomputed using the first 1,349 returns (the subsample containing data before the 1987 crash). We can observe the typical finding that the autocorrelations of r_t^2 are much higher than those of r_t. The transformed series $\log r_t^2$ is significantly autocorrelated both in the full-sample period and in the subsample period. This reveals the kind of nonlinearity in the data that can be explained by the SV model. Strong persistence in the correlations of $\log r_t^2$ probably indicates the presence of strong persistence in volatility and that the series $\log r_t^2$ is close to nonstationary. To filter out the autocorrelations in r_t, the series is fitted by an AR(1) process. The residuals are then extracted for further analysis. An alternative approach is to simulate from the posterior densities of the intercept and the autoregressive parameter via Gibbs sampling. This can be easily performed by specifying Gaussian priors for the two additional parameters. We expect that autocorrelations in the returns have little information on the volatility. Therefore, the former method as used by Jacquier et al. (1994) is adopted. The residual return at time t is denoted by y_t in the following discussion.

6.2 Fitting the SV Model

We perform the Gibbs sampling as given by Kim et al. (in press) on the S & P 500 data. The results are summarized in Table 3, page 247. A typical finding is that the posterior mean of ϕ is close to 1. The half-life of a shock to volatility has a posterior mean of about 21 weeks. A particular shock to volatility takes more than four months for its effect to halve. The posterior 90% interval of ϕ indi-

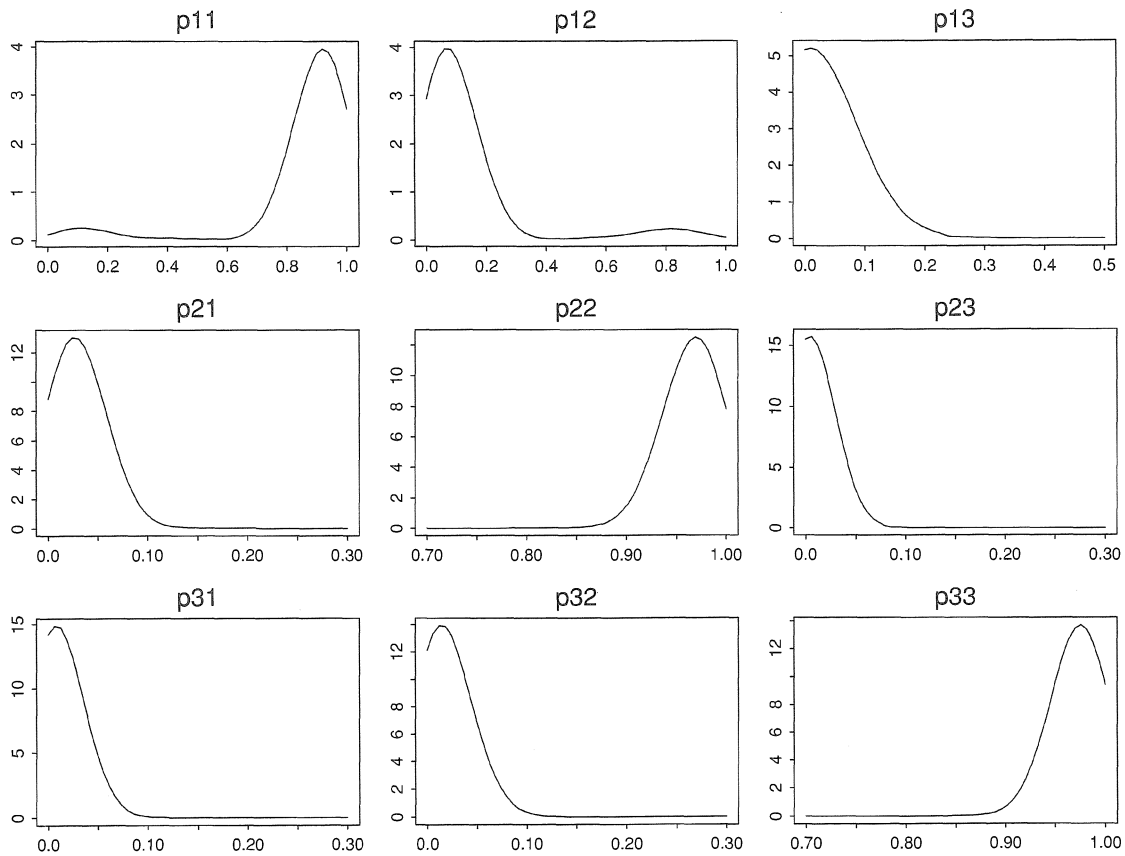


Figure 3. Posterior Densities for the Transition Probabilities for the S & P 500 data in the MSSV(3) Fitting.

cates that persistence of volatility is high. This, of course, is a common conclusion in the ARCH and the SV literature. Figure 1, page 248, plots the marginal posterior densities of α, σ^2, ϕ, and the half-life $(-\log 2/\log \phi)$. The distribution of α is roughly symmetric, whereas the others are asymmetric. Moreover, the distribution of the half-life is very diffuse and skewed to the right, though the posterior mode is close to the posterior mean.

6.3 Fitting the MSSV Model

We consider a three-state MSSV Model for the S & P 500 data. The low-, medium-, and high-volatility states correspond to $s_t = 3, s_t = 2$, and $s_t = 1$, respectively. In implementing the Gibbs sampler, the first $M = 50{,}000$ burn-in iterations are discarded and the subsequent $N = 50{,}000$ iterations are regarded as a simulated sample from the posterior density. The posterior means, posterior standard deviations, and posterior 90% intervals are displayed in Table 4, page 248. The posterior intervals of γ_2 and γ_3 indicate that the two parameters differ from 0. The persistent parameter ϕ has posterior mean equal to .472. Indeed, a similar parameter was estimated to be .48 by Hamilton and Susmel (1994) when analyzing weekly New York Stock Exchange returns. The persistence in volatility as reflected in the SV fitting is resolved by allowing sudden discrete shifts in the intercept. This is a common finding in the work of Hamilton and Susmel (1994) and Cai (1994), in which conditional heteroscedasticity in each regime is accounted for by ARCH disturbances.

The two diagonal entries p_{22} and p_{33} have posterior means close to unity, implying that states 2 and 3 tend to persist for a long time. On average, state 2 and state 3 would last for about 49 and 65 weeks, respectively. Duration in the low-volatility state is longer than that in the medium-volatility state. Once the high-volatility state is achieved, it has about a 13% chance to switch to the medium-volatility state. Persistence in the high-volatility state is relatively low, the mean duration being less than 18 weeks. Moreover, the posterior means of p_{23} and p_{31} are relatively small. This implies that there is little chance for the switching from state 2 to state 3 and from state 3 to state 1. Figures 2 and 3 plot the marginal posterior densities for the unknown parameters. As in Section 5, the plots are created by using Gaussian kernel density estimation. Many of the distributions are asymmetric. The parameters γ_2 and γ_3, which define the three regimes, have distributions that cluster away from 0, indicating that the volatility in the three regimes is substantially different.

Figure 4 displays plots of the returns series and the posterior probabilities $\Pr(s_t = j), j = 1, 2, 3$, for our sample period. Most of the episodes belong to the medium-volatility regime. If we classify the state of a particular time point using the maximal posterior probability, then the number of weeks in states 1, 2, and 3 are 226, 1,017, and 165, respectively. Several spikes with posterior probability greater

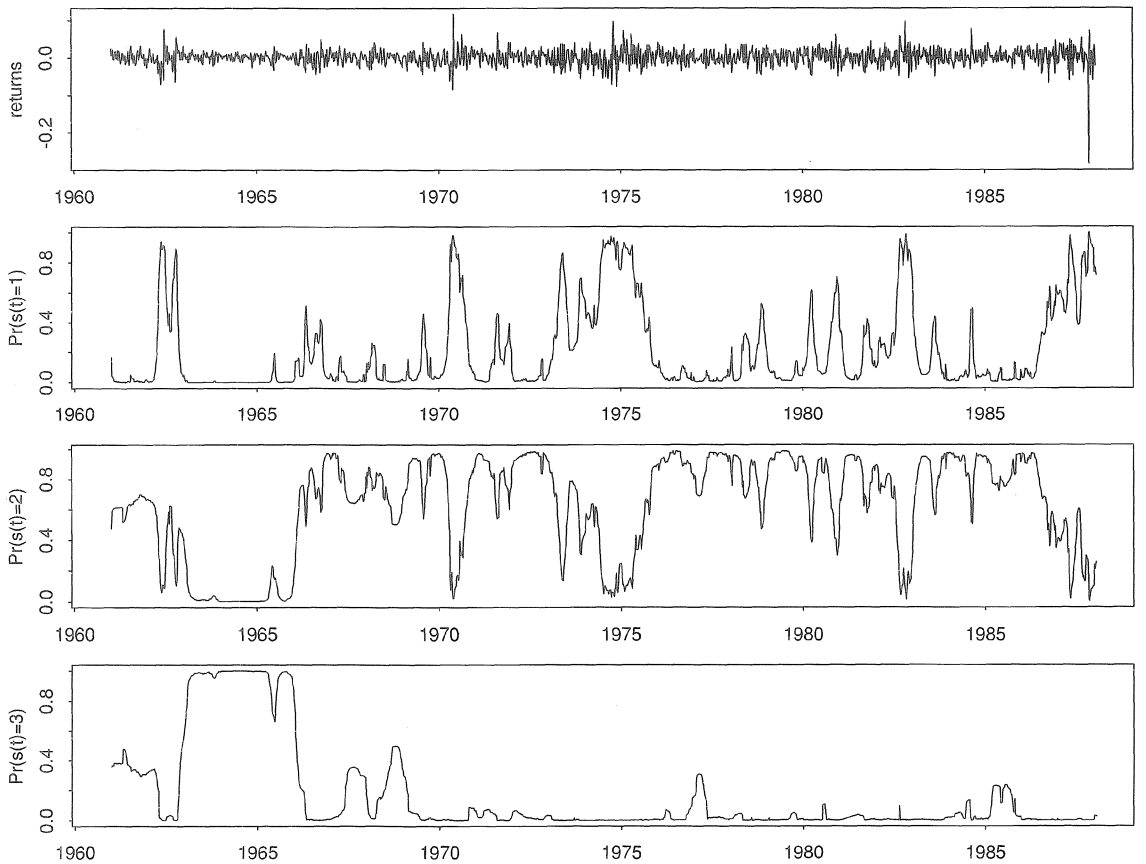


Figure 4. The Returns and the Posterior Probabilities $Pr(s_t = j)$, $j = 1, 2, 3$, for the S & P 500 Data From 1961 to 1987.

than .5 can be observed in the plot for the high-volatility regime. Compared with the time series plot of the returns, the spikes are coincident with the abnormal fluctuations in the series located around the years 1962, 1970, 1975, 1982, 1985, and 1987. The most distinguishable one with posterior probability closest to 1 corresponds to the 1987 crash. Therefore, the episodes associated with the abnormal fluctuations of the stock-market index are closely related to the high-volatility state. Hamilton and Susmel (1994) pointed out that the periods of high stock-market volatility seem to be related to general business downturns. The four U.S. economic recessions mentioned by Hamilton and Susmel (1994) overlap considerably with the periods in the high-volatility regime. Moreover, the quiet period (1962–1965) identified by Hamilton and Susmel (1994) is a subset of our third regime. Though the regimes established in this article using the S & P 500 index are different from that by the SWARCH model using the New York Stock Exchange index, their similarities cannot be ignored.

7. CONCLUSION

This article introduces a new approach in modeling the complex structure of financial time series. The SV model, a natural alternative to the popular ARCH models, is combined with the Markov switching model. The MSSV model enables the intercept coefficient to switch between states so as to capture the effect accompanied with some sudden important events. Traditional estimation problems of fitting the stochastic volatility model are resolved by the MCMC and the data-augmentation methodologies. Samples from the joint posterior distribution of the unknown parameters, as well as the latent variables, are generated by means of Gibbs sampling. Inferences on parameters or latent variables can be carried out effectively by marginalization.

We applied the techniques to the U.S. S & P 500 weekly returns for the years 1961 to 1987. Persistence in volatility, which is a common phenomenon in the ARCH and SV literature, is explained by the persistence in the low- and medium-volatility states. This particular finding and those of Hamilton and Susmel (1994) and Cai (1994) are analogous to each other. Three regimes have been identified for the S & P 500 data. The high-volatility state is closely related to sudden abnormal fluctuations such as the 1987 crash. The high-volatility state covers most of the four U.S. recession periods mentioned by Hamilton and Susmel (1994). Our analysis is, to a certain degree, consistent with the earlier study of the New York Stock Exchange data by Hamilton and Susmel (1994).

Further research may consider the model choice between the SV model and the MSSV model. Methods suggested by Chib (1995) may be used to perform a Bayesian comparison of the two models. The SV model as a discrete-time approximation of some diffusion models is now generalized to per-

mit multiple volatility regimes. Thus, the MSSV model implicitly characterizes discrete levels for option price. There are many factors determining option price, however, so the actual consequence of the volatility regime switching on option pricing needs further research. In short, this more realistic model is expected to be beneficial to option pricing.

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APPENDIX: THE FULL CONDITIONAL DISTRIBUTIONS FOR THE MODEL PARAMETERS

A.1 Full Conditional Distribution of σ^2, γ_i ($i = 1, \ldots, K$) and ϕ

The prior distributions used are $\sigma^2 \sim \mathrm{IG}(\nu_1, \nu_2), \gamma_i \sim N(\bar{\gamma}_i, \delta_i^2)(i = 1, \ldots, K)$, and $\phi \sim N(\bar{\phi}, \beta^2)$, where IG denotes the inverse gamma distribution and $\nu_1, \nu_2, \bar{\gamma}_i, \delta_i^2, \bar{\phi}$, and β^2 are the prespecified hyperparameters. Diffuse priors for the γ_i's, σ^2, and $\phi(\bar{\gamma}_i = -2, \delta_i = 10^6, \nu_1 = 2 + 10^{-20}, \nu_2 = 100, \bar{\phi} = 0$, and $\beta = 10^6)$ were used in the MSSV model. Note that the priors for $\gamma_i, i = 2, \ldots, K$, and ϕ are truncated according to assumptions 1 and 3. According to (7), the conditional posterior distribution of σ^2 is $\mathrm{IG}((n/2) + \nu_1, (1/b))$ where

$$b = \frac{1}{2}\left\{\sum_{t=2}^{n}(\log h_t - \alpha_{s_t} - \phi \log h_{t-1})^2 + (1-\phi^2)\left(\log h_1 - \frac{\alpha_{s_1}}{1-\phi}\right)^2\right\} + \frac{1}{\nu_2}.$$

Define, for each i from 1 to K, time series z_{it} by

$$z_{i1} = \frac{\sqrt{1-\phi^2}}{1-\phi}[(1-\phi)\log h_1 - \alpha_{s_1} + \gamma_i]I_{i1}$$

and

$$z_{it} = (\log h_t - \alpha_{s_t} + \gamma_i - \phi \log h_{t-1})I_{it}, \qquad t = 2, \ldots, n.$$

Then, we have the posterior density of γ_i given by $N(\tilde{\gamma}_i, \tilde{\delta}_i^2)$, where

$$\tilde{\delta}_i^2 = \frac{\delta_i^2\sigma^2(1-\phi)}{\delta_i^2(1+\phi) + (1-\phi)[\sigma^2 + (n-1)\delta_i^2]},$$

$$\tilde{\gamma}_i = \frac{\tilde{\delta}_i^2}{\sigma^2}\left(\frac{\sqrt{1-\phi^2}}{1-\phi}z_{i1} + \sum_{t=2}^{n} z_{it} + \frac{\bar{\gamma}_i\sigma^2}{\delta_i^2}\right),$$

$$J_1 = 1,$$

and

$$J_i = \begin{cases} 1 & \text{if } \gamma_i < 0 \\ 0 & \text{otherwise,} \end{cases} \qquad i = 2, \ldots, K.$$

Similarly, we have the following result for ϕ:

$$f(\phi|\omega_{-\phi}, Y_n) \propto Q(\phi)\frac{1}{\tilde{\beta}}\exp\left[-\frac{1}{2\tilde{\beta}^2}(\phi - \tilde{\phi})^2\right] I(|\phi| < 1),$$

where

$$Q(\phi) = \sqrt{1-\phi^2}\exp\left\{-\frac{1}{2\sigma^2}(1-\phi^2)\left(\log h_1 - \frac{\alpha_{s_1}}{1-\phi}\right)^2\right\},$$

$$\tilde{\beta}^2 = \frac{\sigma^2\beta^2}{\beta^2\sum_{t=2}^{n}(\log h_{t-1})^2 + \sigma^2},$$

$$\tilde{\phi} = \tilde{\beta}^2\left[\frac{\sum_{t=2}^{n}(\log h_t - \alpha_{s_t})\log h_{t-1}}{\sigma^2} + \frac{\bar{\phi}}{\beta^2}\right],$$

and $I(\cdot)$ is an indicator function. By ignoring the $Q(\phi)$ as given by Chib and Greenberg (1994) and Albert and Chib (1993), we can simply draw ϕ from the truncated $N(\tilde{\phi}, \tilde{\beta}^2)$ with the constraint $|\phi| < 1$ using the inverse cdf method.

A.2 Full Conditional Distribution of p_{ij} and π_i ($i, j = 1, \ldots, K$)

Given $i = i^*, p_{i^*j}$ $(j = 1, \ldots, K)$ are probabilities that sum to unity. With the use of the noninformative prior, the full conditional distribution of $(p_{i^*1}, p_{i^*2}, \ldots, p_{i^*K})$ is the Dirichlet distribution $\mathrm{Dir}(d_{i^*1}, d_{i^*2}, \ldots, d_{i^*K})$, where

$$d_{i^*j} = \sum_{t=2}^{n} I(s_{t-1} = i^*)I(s_t = j) + .5.$$

In what follows, $(\pi_1, \ldots, \pi_K)$ are drawn from $\mathrm{Dir}(.5 + I(s_1 = 1), .5 + I(s_1 = 2), \ldots, .5 + I(s_1 = K))$ by specifying the noninformative prior.

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REFERENCES

Albert, J. H., and Chib, S. (1993), "Bayes Inference via Gibbs Sampling of Autoregressive Time Series Subject to Markov Mean and Variance Shifts," *Journal of Business & Economic Statistics*, 11, 1–15.

Andersen, T. G. (1994), "Stochastic Autoregressive Volatility: A Framework for Volatility Modelling," *Mathematical Finance*, 4, 75–102.

Andersen, T. G., and Sørensen, B. (1996), "GMM Estimation of a Stochastic Volatility Model: A Monte Carlo Study," *Journal of Business & Economic Statistics*, 14, 328–352.

Andrews, D. W. K. (1991), "Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation," *Econometrica*, 59, 817–858.

Bollerslev, T., Chou, R. Y., and Kroner, K. F. (1992), "ARCH Modeling in Finance: A Selective Review of the Theory and Empirical Evidence," *Journal of Econometrics*, 52, 5–59.

Bollerslev, T., Engle, R. F., and Nelson, D. B. (1994), "ARCH Models," in *Handbook of Econometrics, Vol. 4*, eds. R. Engle and D. McFadden, New York: North-Holland, pp. 2959–3038.

Cai, J. (1994), "A Markov Model of Switching-Regime ARCH," *Journal of Business & Economic Statistics*, 12, 309–316.

Carlin, B. P., Polson, N. G., and Stoffer, D. S. (1992), "A Monte Carlo Approach to Nonnormal and Nonlinear State-Space Modeling," *Journal of the American Statistical Association*, 87, 493–500.

Carter, C. K., and Kohn, R. (1994), "On Gibbs Sampling for State Space Models," *Biometrika*, 81, 541–553.

Chib, S. (1993), "Bayes Regression With Autoregressive Errors: A Gibbs Sampling Approach," *Journal of Econometrics*, 58, 275–294.

——— (1995), "Marginal Likelihood From the Gibbs Output," *Journal of the American Statistical Association*, 90, 1313–1321.

——— (1996), "Calculating Posterior Distributions and Modal Estimates in Markov Mixture Models," *Journal of Econometrics*, 75, 79–97.

Chib, S., and Greenberg, E. (1994), "Bayes Inference in Regression Models With ARMA(p, q) Errors," *Journal of Econometrics*, 64, 183–206.

Chou, R. Y. (1988), "Volatility Persistence and Stock Valuations: Some Empirical Evidence Using GARCH," *Journal of Applied Econometrics*, 3, 279–294.

Clark, P. K. (1973), "A Subordinated Stochastic Process Model With Finite Variance for Speculative Prices," *Econometrica*, 41, 135–155.

De Jong, P., and Shephard, N. (1995), "The Simulation Smoother for Time Series Models," *Biometrika*, 82, 339–350.

Engle, R. F. (1982), "Autoregressive Conditional Heteroscedasticity With Estimates of the Variance of United Kingdom Inflation," *Econometrica*, 50, 987–1008.

French, K. R., Schwert, G. W., and Stambaugh, R. F. (1987), "Expected Stock Returns and Volatility," *Journal of Financial Economics*, 19, 3–30.

Gallant, A. R., Hsieh, D. A., and Tauchen, G. E. (1991), "On Fitting a Recalcitrant Series: The Pound/Dollar Exchange Rate, 1974–83," in *Nonparametric and Semiparametric Methods in Econometrics and Statistics*, eds. W. A. Barnett, J. Powell, and G. E. Tauchen, Cambridge, U.K.: Cambridge University Press, pp. 199–240.

Gelfand, A. E., Hills, S. E., Poon, A. R., and Smith, A. F. M. (1990), "Illustration of Bayesian Inference in Normal Data Models Using Gibbs Sampling," *Journal of the American Statistical Association*, 85, 972–985.

Gelfand, A. E., and Smith, A. F. M. (1990), "Sampling-Based Approaches to Calculating Marginal Densities," *Journal of the American Statistical Association*, 85, 398–409.

Geman, S., and Geman, D. (1984), "Stochastic Relaxation, Gibbs Distribution and the Bayesian Restoration of Images," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 6, 721–741.

Geweke, J. (1994), Comment on "Bayesian Analysis of Stochastic Volatility Models," *Journal of Business & Economic Statistics*, 12, 397–399.

Ghysels, E., Harvey, A. C., and Renault, E. (in press), "Stochastic Volatility," in *Statistical Methods in Finance*, Amsterdam: North-Holland.

Hamilton, J. D. (1988), "Rational-Expectations Econometric Analysis of Changes in Regime: An Investigation of the Term Structure of Interest Rates," *Journal of Economic Dynamics and Control*, 12, 385–423.

——— (1989), "A New Approach to the Economic Analysis of Nonstationary Time Series and the Business Cycle," *Econometrica*, 57, 357–384.

Hamilton, J. D., and Susmel, R. (1994), "Autoregressive Conditional Heteroskedasticity and Changes in Regime," *Journal of Econometrics*, 64, 307–333.

Hansen, B. E. (1992), "Consistent Covariance Matrix Estimation for Dependent Heterogeneous Processes," *Econometrica*, 60, 967–972.

Harris, L. (1987), "Transaction Data Tests of the Mixture of Distributions Hypothesis," *Journal of Financial and Quantitative Analysis*, 22, 127–141.

Harvey, A. C. (1993), *Time Series Models*, New York: Wiley.

Harvey, A. C., Ruiz, E., and Shephard, N. (1994), "Multivariate Stochastic Variance Models," *Review of Economic Studies*, 61, 247–264.

Harvey, A. C., and Shephard, N. (1996), "Estimation of an Asymmetric Stochastic Variance Model for Asset Returns," *Journal of Business & Economic Statistics*, 14, 429–434.

Heston, S. L. (1993), "A Closed-Form Solution for Options With Stochastic Volatility With Applications to Bond and Currency Options," *Review of Financial Studies*, 6, 327–343.

Hull, J., and White, A. (1987a), "The Pricing of Options on Assets With Stochastic Volatilities," *Journal of Finance*, 42, 281–300.

——— (1987b), "Hedging the Risks From Writing Foreign Currency Options," *Journal of International Money and Finance*, 6, 131–152.

Jacquier, E., Polson, N. G., and Rossi, P. E. (1994), "Bayesian Analysis of Stochastic Volatility Models," *Journal of Business & Economic Statistics*, 12, 371–389.

Kim, S., Shephard, N., and Chib, S. (in press), "Stochastic Volatility: Likelihood Inference and Comparison With ARCH Models," *Review of Economic Studies*, 65.

Lamoureux, C. G., and Lastrapes, W. D. (1990), "Persistence in Variance, Structural Change, and the GARCH Model," *Journal of Business & Economic Statistics*, 8, 225–234.

Metropolis, N., Rosenbluth, A. W., Rosenbluth, M. N., Teller, A. H., and Teller, E. (1953), "Equations of State Calculations by Fast Computing Machines," *Journal of Chemical Physics*, 21, 1087–1092.

Nelson, D. B. (1988), "Time Series Behavior of Stock Market Volatility and Returns," unpublished Ph.D. dissertation, Massachusetts Institute of Technology.

Poon, S. H., and Taylor, S. J. (1992), "Stock Returns and Volatility: An Empirical Study of U.K. Stock Market," *Journal of Banking and Finance*, 16, 37–59.

Ripley, B. D. (1987), *Stochastic Simulation*, New York: Wiley.

Shephard, N. (1993), "Fitting Nonlinear Time-Series Models With Applications to Stochastic Variance Models," *Journal of Applied Econometrics*, 8, S135–S152.

——— (1994), "Partial Non-Gaussian State Space," *Biometrika*, 81, 115–131.

——— (1996), "Statistical Aspect of ARCH and Stochastic Volatility," in *Time Series Models: In Econometrics, Finance and Other Fields*, eds. D. R. Cox, D. V. Hinkley, and O. Barndorff-Nielsen, London: Chapman & Hall, pp. 1–67.

Shephard, N., and Kim, S. (1994), Comment on "Bayesian Analysis of Stochastic Volatility Models," *Journal of Business & Economic Statistics*, 12, 406–410.

So, M. K. P., Lam, K., and Li, W. K. (1997), "An Empirical Study of Volatility in Seven Southeast Asian Stock Markets Using ARV Models," *Journal of Business Finance & Accounting*, 24, 261–275.

So, M. K. P., Li, W. K., and Lam, K. (1997), "Multivariate Modelling of the Autoregressive Random Variance Process," *Journal of Time Series Analysis*, 18, 429–446.

Tanner, M. A., and Wong, W. H. (1987), "The Calculation of Posterior Distribution by Data Augmentation," *Journal of the American Statistical Association*, 82, 528–540.

Tauchen, G. E., and Pitts, M. (1983), "The Price Variability–Volume Relationship on Speculative Markets," *Econometrica*, 51, 485–505.

Taylor, S. J. (1982), "Financial Returns Modelled by the Product of Two Stochastic Processes, a Study of Daily Sugar Prices 1961–79," in *Time Series Analysis: Theory and Practice 1*, ed. O. D. Anderson, Amsterdam: North-Holland, pp. 203–226.

——— (1986), *Modelling Financial Time Series*, New York: Wiley.

——— (1994), "Modeling Stochastic Volatility," *Mathematical Finance*, 4, 183–204.

Tierney, L. (1994), "Markov Chains for Exploring Posterior Distributions" (with discussion), *The Annals of Statistics*, 21, 1701–1762.

Wiggins, J. B. (1987), "Option Values Under Stochastic Volatility: Theory and Empirical Estimates," *Journal of Financial Economics*, 19, 351–372.